

Conduction vs Insulation by RealQM

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Abstract

This article shows what RealQM, as a new alternative to textbook quantum mechanics, offers as explanation of the difference between conductors and insulators. RealQM represents an N -electron system by N non-negative unit charge densities in ordinary three-dimensional space, one per electron, meeting at free boundaries and minimising the total energy — Coulomb interaction plus kinetic energy, with no other force and nothing fitted. It is a computable, parameter-free model at polynomial cost.

Computationally the difficulty is one of scale. Atomic energies are tens of electron-volts; semiconductor activation is 50–500 meV; metallic conduction sets in near 20 meV. Against the total energy of a system large enough for the question to be posed those are 10^{-3} , 10^{-5} and 10^{-6} , so whether a material conducts is decided by a difference five or six orders of magnitude below the number a calculation returns. The same span falls on the model: a description adequate for binding, where a per cent error is invisible, can be hopeless for conduction, where a per cent of the total is a hundred times the quantity in question.

Two models are measured. A **chain of atoms**, the valence electron occupying its own kernel's region, gives a corrugation of order an electron-volt for the electron pattern to slide through, and an extensive polarisability $\alpha = 27.4N$: an insulator. A **cable of ions surrounded by electrons**, the carriers standing off cores that screen the kernels, gives ≈ 240 meV — the semiconductor range, where transport is thermally activated hopping. Raising the coefficient of the kinetic energy from the $\frac{1}{2}$ of atomic standardisation to 1, which compensates for a free interface that resists neither compression nor modulation, gives ≈ 24 meV: the pinning no longer holds and charge is free to move.

The corrugation is the activation energy E_a , fixing the exponential in the hopping conductivity $\sigma \propto \nu \exp(-E_a/kT)$, so 240 against 24 meV is four orders of magnitude in σ at room temperature. The prefactor ν is an attempt frequency, a lattice quantity this framework can also supply, though it is not computed here. Metallic conduction is different: its rate needs a relaxation time from electron-phonon scattering, and that stays outside.

1 The framework, and what is asked of it

RealQM represents an N -electron system by N non-negative densities ψ_i , each carrying unit charge on its own domain Ω_i , the domains non-overlapping and their interfaces free to move to where the energy is least. The energy minimised is

$$E[\{\psi_i\}, \{\Omega_i\}] = \sum_i \int_{\Omega_i} \left(\frac{1}{2} |\nabla \psi_i|^2 - \sum_a \frac{Z_a}{|x - x_a|} \psi_i^2 \right) dx + \frac{1}{2} \sum_{i \neq j} \iint \frac{\psi_i^2(x) \psi_j^2(y)}{|x - y|} dx dy, \quad (1)$$

subject to $\int_{\Omega_i} \psi_i^2 = 1$. There is no wavefunction in $3N$ dimensions, no antisymmetry and no self-interaction; the role played by the exclusion principle is played by the requirement that domains do

not overlap. The minimisation runs over the partition as well as over the densities, so the domain edges are unknowns of the problem. Everything is Coulomb interaction and kinetic energy, with nothing fitted, and the cost is polynomial in N rather than exponential.

Every previous test of this construction — ionisation energies across the periodic table, molecular binding, hydrogen bonds and solvation, nuclear binding by dual Coulomb confinement — has been on *bound* matter, where an attractor is always present and the charge is where the attractor is. Conduction is the first question that asks about charge which is not where an attractor holds it, and the framework has no separate machinery for it: the same energy, the same domains, in a different arrangement of the same two kinds of charge.

Two arrangements, both ordinary matter. The first is a **chain of atoms**: one kernel per site with its valence electron occupying the same region, which is what a hydrogen chain is and what an alkali is to a first approximation. The second is a **cable of ions surrounded by electrons**: kernels screened by their own core electrons, with the valence charge standing off outside them, which is how a wire is built. The question is what each gives.

2 What is measured

A domain carries one unit of charge by constraint, so no charge passes between sites. What can move is the *partition*: the boundaries are redrawn, and the pattern of domains advances through the lattice. When the pattern has advanced by one full spacing, every electron has taken over its neighbour's kernel and the configuration is restored — identical, since the electrons are identical and the energy depends on the partition and not on labels. Charge has been transported and nothing material has traversed the sample, which is the relation a dislocation bears to slip.

The quantity that decides whether this happens is the energy along that path. Write $E(d)$ for the energy with the partition displaced by a fraction d of a spacing. It is periodic, $E(d+1) = E(d)$, and its amplitude

$$V_0 = \max_d E(d) - \min_d E(d) \quad (2)$$

is the corrugation: the barrier the partition must cross to advance one site. A field tilts this curve by $-NeFd$, so the pattern runs when the tilt exceeds the steepest slope of $E(d)$, and otherwise settles at a displacement and stops — bounded polarisation, no current.

Ends, and how to remove them. A finite chain has ends, and they dominate: terminal domains bind an electron-volt harder than interior ones, so a corrugation read from total energies is reading the ends. Displacing only the middle M carriers repairs this. The energy difference is then M times the bulk cost plus two junction costs, so differencing two window sizes cancels the junctions exactly, and no terminal domain moves in either configuration:

$$\Delta E(M) = M \varepsilon_{\text{bulk}} + 2\varepsilon_{\text{junc}}, \quad \varepsilon_{\text{bulk}} = \frac{\Delta E(M_2) - \Delta E(M_1)}{M_2 - M_1}. \quad (3)$$

The decomposition asserts linearity in M , which a third window tests rather than fits.

3 Model I: a chain of atoms

Seven model atoms in a row, one valence electron each, the electron occupying its kernel's region. Two measurements.

The polarisability is extensive. With the partition held fixed and a uniform field applied, the induced dipole gives α , and repeating at three chain lengths gives its scaling: $\alpha = 27.4N$, flat to 1% per atom. That is the signature of charge fixed per site. A metal’s polarisability grows faster than the number of atoms, because charge redistributes over the whole sample; here each domain carries its unit and cannot give it up, so the only responses are the density deforming inside a fixed domain and the boundaries displacing. Both are per-atom constants, and the sum is exactly proportional to N . The one escape is a vanishing corrugation, which would let the partition slide freely and the response grow without bound.

The corrugation does not vanish. It is of order an electron-volt per electron, and it stays there: repeating the measurement across lattice spacings from 3.5 to $8.0 a_0$ gives a minimum near the equilibrium spacing and a rise on both sides, with no spacing approaching zero and no flip of the registry preference. Compressed, the cores crowd the electron; expanded, sharing a kernel costs more than owning one.

An electron-volt is forty times thermal energy at room temperature and lies above the 10–500 meV of the activated conductors this framework might otherwise describe. On this model the answer is unambiguous: an insulator, and not a marginal one.

4 Model II: a cable of ions surrounded by electrons

The chain places the carrier where the attractor is. Conducting matter does not: the core electrons occupy the region around each nucleus and the conduction charge stands outside them. Representing that needs two populations — cores held inside a radius, carriers outside it, with only the carriers free to slide — and it is the one arrangement that gives a carrier a standoff without violating the tiling, because the inner region is genuinely owned rather than left empty.

Sixteen sites, kernels of charge $+3$ on the axis, a closed core of two electrons at each, and one valence electron per site outside them: neutral, with no pseudopotential anywhere, so the exclusion of the carrier from the core region is by non-overlap rather than by an imposed condition. The corrugation, measured by (3) with windows of four and eight carriers and checked against a third at twelve:

$$\varepsilon_{\text{bulk}} \approx 240 \text{ meV per carrier.}$$

That is inside the 10–500 meV band of the doped semiconductors, organic conductors and polaronic oxides — the class of materials whose transport is thermally activated hopping, and the class this framework has a claim on. The same interaction, the same energy, the same domains: only the arrangement of the charge has changed, and the barrier has fallen by a factor of four.

5 What sets the pinning

The barrier is not set by the strength of the ion lattice but by whether the carriers bunch into its wells. A carrier distribution that resisted being modulated would ride over the lattice; one that does not resist locks to it.

RealQM’s free interface resists very little, and this is internal to the construction. A flat density is admissible on a free boundary at no cost — a constant ψ_i satisfies the normalisation, has $\nabla\psi_i = 0$, and minimises the electrostatic energy against a neutralising background — so redrawing an interface between domains of equal density costs nothing. The partition therefore offers no resistance to uniform compression, and the only penalty on a *modulated* density is the gradient term in (1), carrying the coefficient $\frac{1}{2}$.

That coefficient is a standardisation, not a constant of the theory. It is set by calibrating the framework against atomic spectra, hydrogen first among them — on a bound electron in a single potential well. An extended arrangement is a different object, as an effective mass in a solid differs from the free-electron mass without anyone regarding it as tampering, and the coefficient governing how a carrier lattice resists modulation by an ion lattice has no obligation to equal the one an atom’s spectrum sets. Raising it is therefore a question rather than a fitted parameter: what stiffness does an extended arrangement carry?

Raised from $\frac{1}{2}$ to 1, with the ions, the standoff, the geometry and the mesh unchanged:

$$\varepsilon_{\text{bulk}} \approx 24 \text{ meV per carrier,}$$

a fall by a factor of eleven, to thermal energy at room temperature. The pinning no longer holds and the charge is free to move — not as a metal, since non-overlap leaves no extended states to fill into bands, but as a charge lattice sliding collectively over the ions.

model	corrugation per carrier	regime
chain of atoms: carrier on its own kernel	$\sim 1 \text{ eV}$	insulator
cable of ions: carriers standing off the cores	$\approx 240 \text{ meV}$	semiconductor: activated hopping
the same cable, carriers stiffened	$\approx 24 \text{ meV}$	metallic onset: unpinned

6 What this gives, and what it does not

In the activated regime the corrugation is the activation energy, and the conductivity of a hopping conductor is $\sigma \propto \nu \exp(-E_a/kT)$. The measurements therefore supply the factor that decides the temperature dependence and the ratio between regimes: 240 against 24 meV is four orders of magnitude in σ at room temperature. What is missing is the prefactor ν , an attempt frequency — the carrier sits in a well and tries once per vibrational period, so $\nu \simeq (1/2\pi)\sqrt{K/M}$ with K the curvature of the same landscape measured here and M an ionic mass. Both are available in this framework, which computes forces on kernels and moves nuclei under classical equations of motion. An absolute hopping conductivity is therefore a phonon calculation away rather than a missing piece of theory. It is not attempted here.

Metallic conduction is different and is outside. There $\sigma = ne^2\tau/m$, with τ a momentum relaxation time set by electron–phonon scattering, which requires a current as an object, electron dynamics, and irreversibility. A minimisation over real densities has none of the three. Non-overlap also leaves no extended states to fill into bands, so band conduction is absent in kind and not by degree, and no choice of stiffness produces it.

7 Status

The corrugations above are rigid-slide values: all carriers displaced together, the pattern translating without deforming. That is the expensive path. The cheap one is a kink — a single carrier advancing while the rest hold still, compressing the domains ahead and expanding those behind — whose cost is independent of chain length, and which is what real conductors use. The kink barrier is not measured here, so every number above is an upper bound, and the true barriers are lower by a factor this article does not determine.

The states carry Euler–Lagrange residuals of $2\text{--}3 \times 10^{-2}$, against 4×10^{-3} for the chain, so they are not fully relaxed; the stiffened runs use a halved timestep and are less relaxed than the others; and the factor of eleven rests on two values of the coefficient rather than a curve. The

chain’s corrugation depends on a pseudopotential exclusion rule that is not settled: excluding a domain from its own core alone under-excludes, excluding from every core over-confines, and the two bracket the answer rather than fixing it. What does not depend on any of this is the instrument. Equation (3) cancels ends and junctions by construction and tests itself against a third window, and the controls throughout are of the kind that can fail: translation by one spacing must return the same energy to the last digit, and domains related by a symmetry must have equal eigenvalues.

8 Conclusion

Asked whether it distinguishes a conductor from an insulator, RealQM answers differently for two arrangements of the same charge. Where the valence electron occupies its own kernel’s region it insulates, with a corrugation of an electron-volt and a polarisability extensive in the number of electrons. Where the carriers stand off ion cores — the arrangement conducting matter actually has — the barrier falls into the range of the activated conductors. And the barrier is set not by the lattice but by how strongly the carrier distribution resists being modulated by it, which is a property of the interface condition rather than of the interaction: a free interface resists nothing, and supplying the missing stiffness removes the pinning.

What the framework then delivers is the activation energy, and with it the temperature dependence of a hopping conductivity and the ratio between regimes. What it does not deliver is an absolute rate: that needs a timescale, which for hopping is a lattice frequency it could compute and has not, and for metallic conduction is a relaxation time it cannot.

A Computational details

All calculations use the RealQM solver `molecule.js` driven by `chain_slide.html` (WebGPU, single GPU). Equation (1) is minimised by imaginary-time relaxation of the densities on a uniform Cartesian grid, with the partition specified by a cell-ownership map. For every corrugation the partition is *frozen*: the energy of a stated configuration is evaluated, which is legitimate for configurations related by a symmetry of the lattice and is why translation by one spacing is used as the control. Energies are reported in hartree and converted at $1 \text{ Ha} = 27211 \text{ meV}$.

Model I: chain of atoms. $N = 7$ model atoms in a row, nuclear charge $+1$, one valence electron each, wrapped labelling so that translation by one spacing is an exact relabelling. Two geometries were used: $r_c = 1.0 a_0$ at spacings $a = 3.5$ to $8.0 a_0$ for the polarisability and the density trend, and $r_c = 1.93 a_0$ at $a = 5.5 a_0$, $h = 0.448 a_0$ for the corrugation, r_c in the second being fixed by the isolated atom’s ionisation energy rather than chosen. Polarisability from the induced dipole at two field strengths with the partition frozen, at $N = 3, 5, 7$: $\alpha = 27.4 N a_0^3$, flat to 1% per atom. Corrugation from $E(0)$ and $E(\frac{1}{2})$; on the $a = 5.5$ chain, $E(0) = -1.631277 \text{ Ha}$ throughout, with $E(\frac{1}{2}) = -1.839 \text{ Ha}$ under own-core exclusion (808 meV per electron) and -1.617 Ha under exclusion from every core (54 meV). These bracket the answer; the electron-volt scale quoted in §3 is the own-core end, and the ~ 0.8 – 1.5 eV range across spacings from 3.5 to $8.0 a_0$ is measured on the $r_c = 1.0$ chain.

Model II: cable of ions surrounded by electrons. $N = 16$ sites at $a = 4.5 a_0$ on the axis, kernel charge $+3$, one core domain per site carrying two electrons and confined within $c_r = 2.4 a_0$ of the axis, one valence electron per site outside that radius: 32 domains, neutral, $r_c = 0$ with the nuclear singularity clamped at an absolute $0.5 a_0$ so the potential is the same object at every mesh.

Mesh 8 cells per spacing, $h = 0.5625 a_0$, box $59.5 a_0$, grid 106^3 . Only the shell slides. Energies at $d = \frac{1}{2}$ with M middle carriers displaced, against the undisplaced reference $E = -160.133661$ Ha:

M	4	8	12
E (Ha)	-160.352807	-160.312817	-160.281810
ΔE (meV)	-5963	-4875	-4031

Equation (3) on (4, 8) gives $\varepsilon_{\text{bulk}} = 272$ meV and $\varepsilon_{\text{junc}} = -3526$ meV per junction; on (8, 12) it gives 211 meV. The linearity test predicts $\Delta E(12) = -3787$ meV against -4031 measured, a 6% miss, and the spread between the two estimates is quoted as the uncertainty: 240 ± 30 meV. The (4, 8) pair is the cleaner, both its junctions lying well inside the chain, where at $M = 12$ only two sites remain on either side.

Stiffened carriers. Identical geometry, mesh and ion lattice, with the coefficient of $|\nabla\psi_i|^2$ raised from $\frac{1}{2}$ to 1 for the shell domains only, the cores keeping $\frac{1}{2}$. The coefficient enters three places — the imaginary-time step, the $H\psi$ diagnostic and the energy sum — and all three must carry it or the reported energy does not belong to the state produced. The explicit step requires the diffusion number below $1/6$, so the timestep is halved in proportion; the stiffened runs are correspondingly less relaxed. $\Delta E(4) = -8447$ meV, $\Delta E(8) = -8352$ meV, giving $\varepsilon_{\text{bulk}} = 24$ meV per carrier.

Controls applied. Translation by one full period must return the same energy: satisfied to nine digits ($E(0) = E(N)$ exactly) on the wrapped geometries. Repeat runs of one configuration agree to nine digits, so the solver is deterministic and any discrepancy is attributable to the model or the input. Domains related by a symmetry must have equal eigenvalues: this is what exposed the terminal domains of a finite cable binding an electron-volt harder than the interior, and hence the necessity of (3). Euler–Lagrange residuals are 4×10^{-3} for the chain and $2\text{--}3 \times 10^{-2}$ for the cable.

References

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